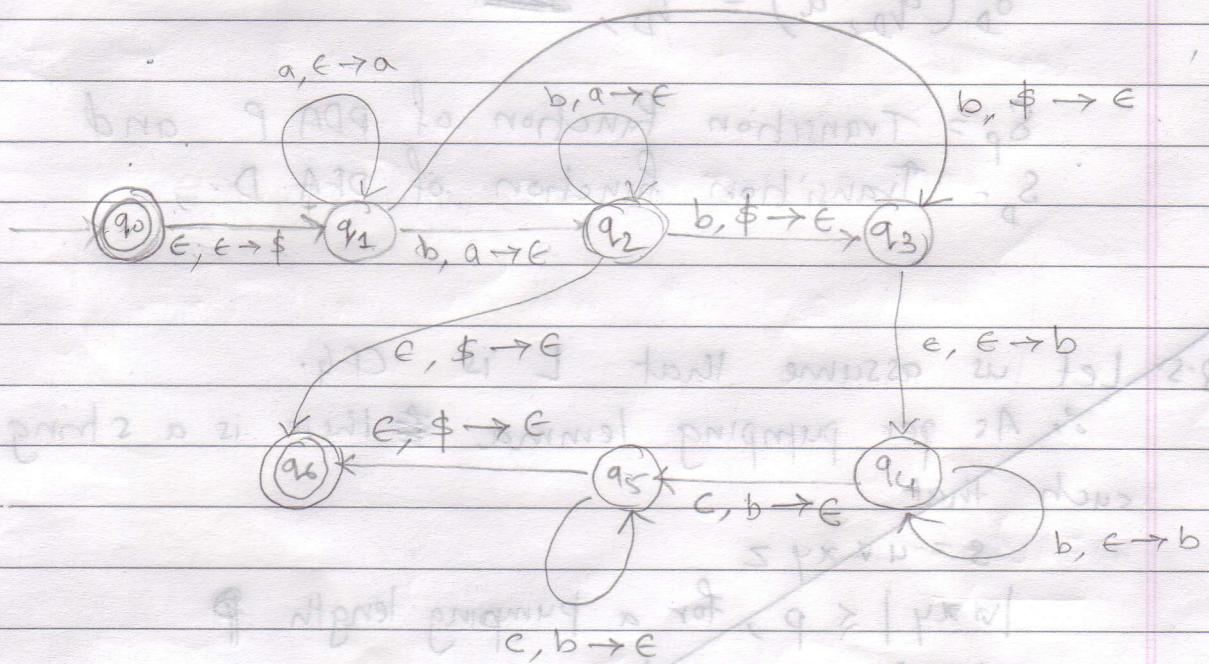


FOUNDATIONS OF COMPUTING THEORY - HOMEWORK 06

Q.2)



Q.3)

$P' = (Q, \Sigma, \Gamma, \delta, q_0, F)$ such that

$Q = Q_P \times Q_D$ where Q_P = states in machine PDA P / DFA D

$\Sigma = \Sigma_P \times \Sigma_D$ where Σ_P = Alphabet for machine PDA P / DFA D

$\Gamma = \Gamma_P$ where Γ_P = stack alphabets of the PDA P

$q_0 = (q_{0P}, q_{0D})$ where q_{0P} = start state of PDA P / DFA D

$F = F_P \times F_D$ where F_P = Final states of PDA P / DFA D

$$\delta((q_P, q_D), \Gamma, a) = \{((q'_P, q'_D), a')\}$$

where $a \in \Sigma_P$, $q_P \in Q_P$, $q_D \in Q_D$ such that $a' \in \Gamma_P$

$$\delta_p(q_p, \Gamma, a) = \{(q'_p, a')\} \text{ and}$$

$$\delta_D(q_D, a) = q'_D, \text{ and}$$

δ_p = Transition function of PDA P and
 δ_D = Transition function of DFA D.

Q.5 Let us assume that L is CFG.

As per pumping lemma there is a string S such that

$$S = uvxyz$$

$$|vxy| \leq p, \text{ for a pumping length } p$$

$$|vy| \geq 1 \text{ and}$$

$$uv^i xy^i z \in L$$

let us consider $S = a^{p^2}$

$$|a^{p^2}| > p \text{ and } S \in L$$

Q.5 Let us assume that L is CFG.

According to pumping lemma for CFG's, there is a number p such that for any $S \in L$ with $|S| \geq p$, then $S = uvxyz$

such that

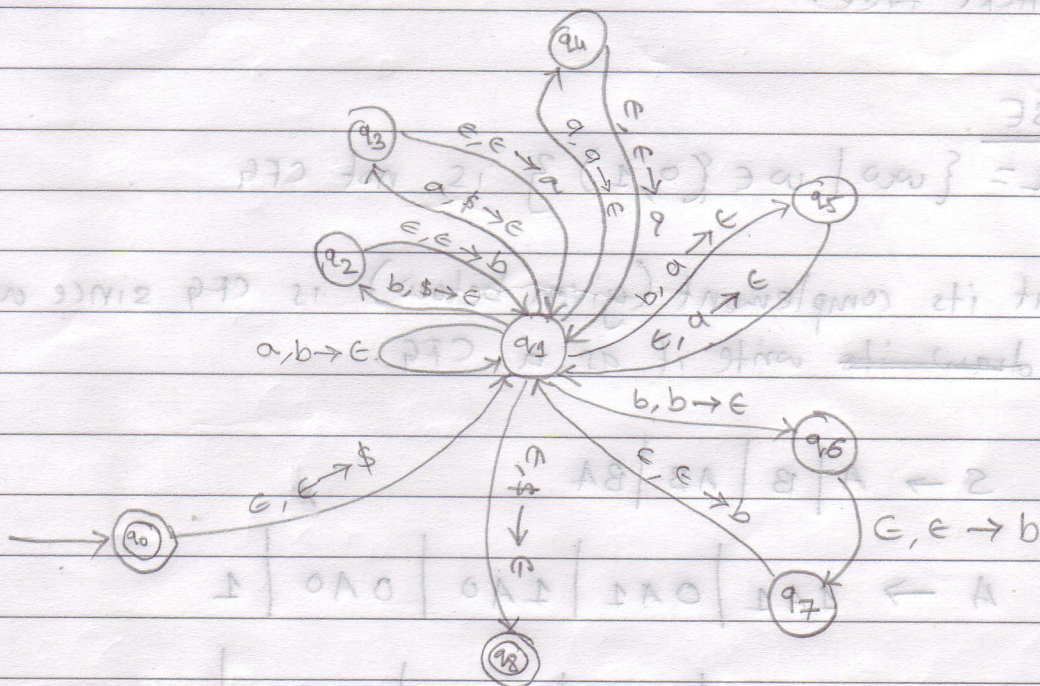
$$uv^i xy^i z \in L \text{ for each } i \geq 0$$

$$|vy| \geq 1 \text{ and}$$

$$|vxy| \leq p$$

- Let us consider $s = a^{p^2} \in L$ and ~~a^{p^2}~~
- Now $|vy| > 0$ so at least one of them needs to have an a . So the $\min(|vy|) = 1$.
- ~~Also~~ Also, $|vxy| \leq p$ so the $\max(|vxy|) \leq p$.
So the ~~$\max(|vy|)$~~ $\max(|vy|) = p$
- Now if we pump the string s by $i=2$ then we get a 's in the range (p^2+1) to (p^2+p) , all of which are less than $(p+1)^2$ which is the next length of string in the language.
- Hence the pumping lemma does not work on our CFG L . But pumping lemma is true for all CFG's. Therefore, our assumption is wrong.
- Therefore, L is not CFG.

Q.3.



Q.1a) FALSE

Let $L = \{a^i \mid i > 0 \text{ and } i \text{ is prime}\}$ and w is a Palindrome

Let $L_1 = \{a^i \mid i > 0 \text{ and } i \text{ is prime}\}$

$L_2 = \{a^i \mid i > 0 \text{ and } i \text{ is not prime}\}$

$L_1 \cup L_2 = a^+$ which is a CFG and regular

b) FALSE

Let $L_1 = \{a^i b^j c^i \mid i > 0\}$

$L_2 = \{a^i b^j c^i \mid 1 < i < j < 2i\}$

then $L_1 \cap L_2 = \emptyset$ which is regular and context free

c) FALSE

$L = \{ww \mid w \in \{0,1\}^*\}$ is not CFG

But its complement (given below) is CFG since we can draw its write it as a CFG.

$S \rightarrow A \mid B \mid AB \mid BA$

$A \rightarrow 1A1 \mid 0A1 \mid 1A0 \mid 0A0 \mid 1$

$B \rightarrow 0B0 \mid 1B0 \mid 0B1 \mid 1B1 \mid 0$